# Numerical Exercise #2

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# 1 Question #1

Numerical solution for the bare system
keff (scientific format with 5 decimal points):
$9.71470 \times 10^{-1}$ [-]
net current at the core boundary (scientific format with 5 decimal points):
$7.55662 \times 10^{-2}\ \mathrm{n\,cm^{-2}\,s^{-1}}$

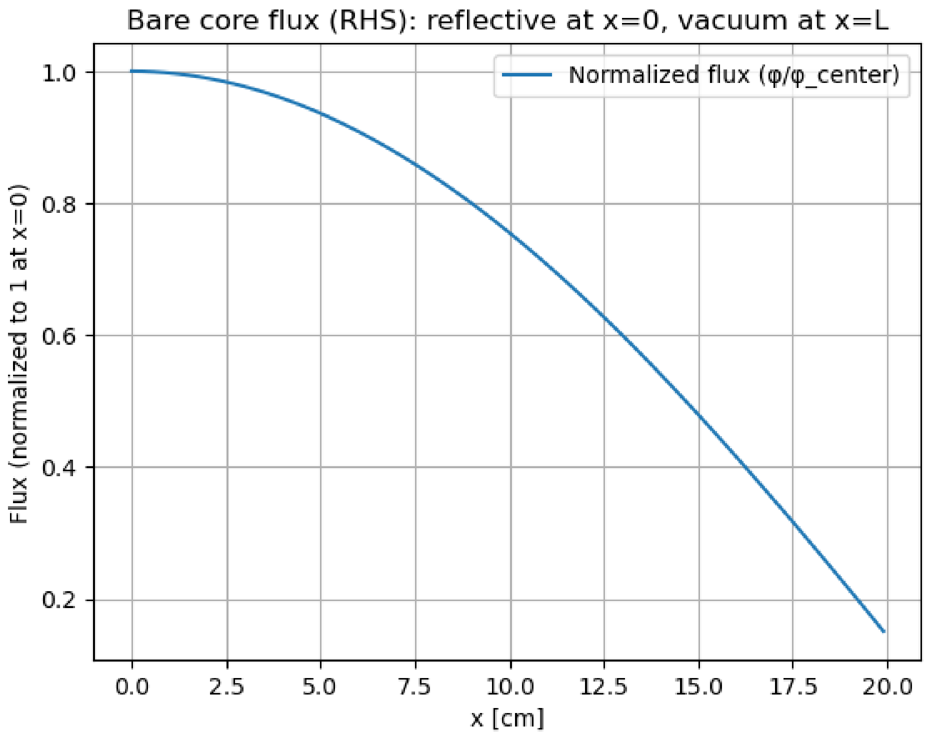


Figure 1: Flux in the bare reactor for a mesh size of 0.1 cm.

# 2 Question #2

Analytical solution for the bare system
keff (scientific format with 5 decimal points):
$9.73560 \times 10^{-1}$ [-]
net current at the core boundary (scientific format with 5 decimal points):
$7.52832 \times 10^{-2}\ \mathrm{n\,cm^{-2}\,s^{-1}}$

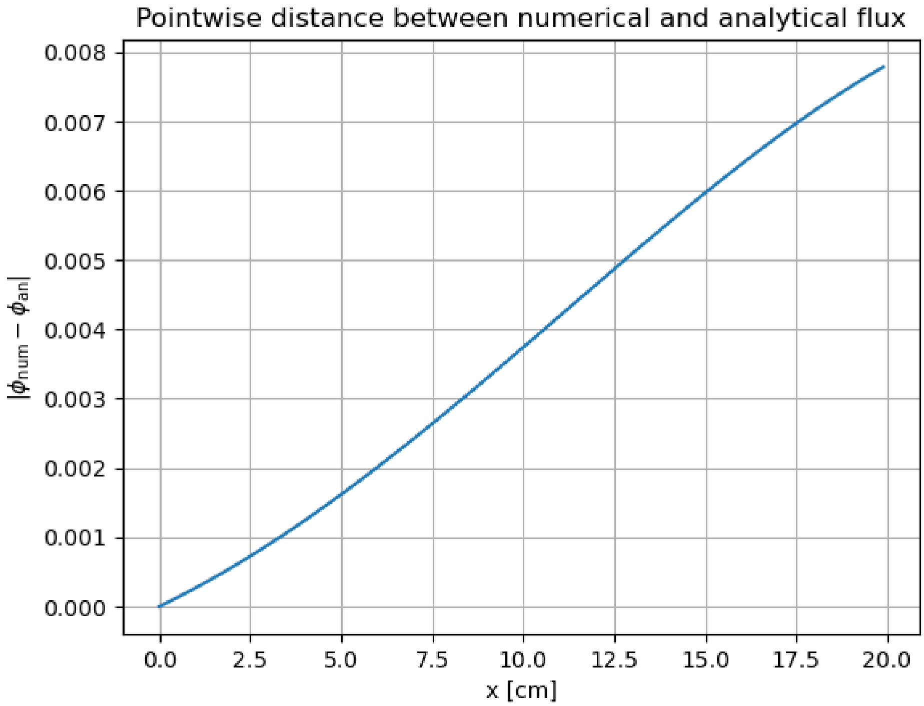


Figure 2: Distance between the solutions at each mesh point for a mesh size of 0.1 cm.

# 3 Question #3

Numerical solution for the reflected system
keff (scientific format with 5 decimal points):
1.120 12 [-]
net current at the core boundary (scientific format with 5 decimal points):
$5.443\,99 \times 10^{-2}$ $\mathrm{n\,cm^{-2}\,s^{-1}}$

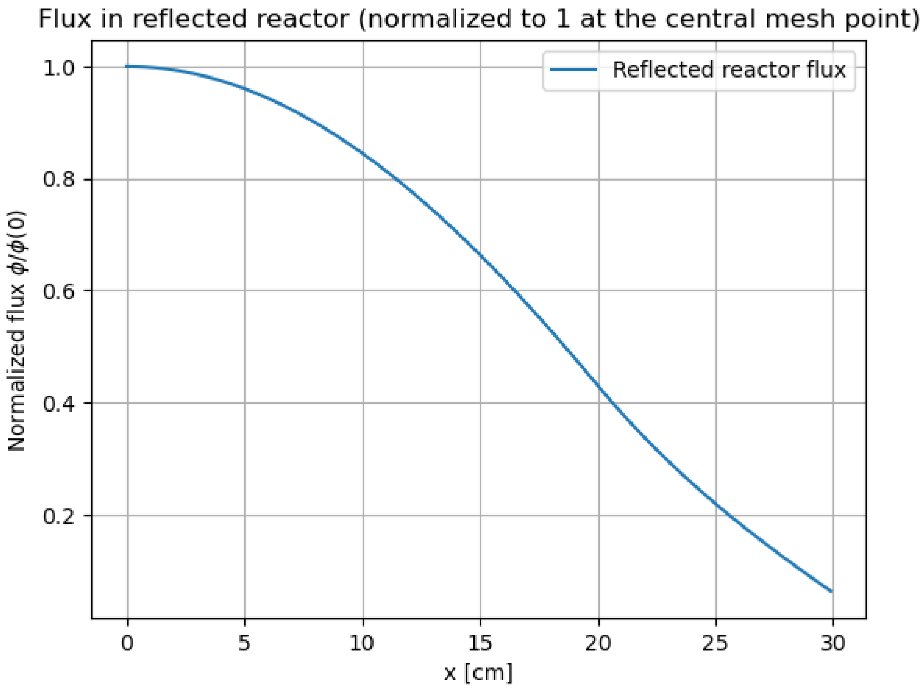


Figure 3: Flux in the reflected reactor for a mesh size of 0.1 cm.